Design Brief – Review 2

0. Header

Project code	FA26SE260
Project name	CampusStay FPT – Dormitory Registration and Room-Scheduling Management System for FPT University's Tân Uyên Campus, Binh Duong
Team	CampusStay FPT — Lê Thành Minh Trung (Leader, SE183752), Nguyễn Xuân Lộc (SE183782), Trần Võ Quang Duy (SE182146), Võ Minh Luân (SE182794)
Supervisor	Thân Thị Ngọc Vân
Repository link (tag review-2)	https://github.com/tisou47/CampusStay-FPT/tree/review-2
Version	1.1
Date	30/10/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
Team, context, actors	No members; Bình Dương; Student, Staff, Admin	4 members; Tân Uyên ward, HCMC; basis of requirements stated; + Manager (all Blocks), Staff per Block	Review 1 feedback; 2025 merger
Room choice, contract, check-in/out	Select room/bed; digital contract; self check-in/out	Choose Room Type and Class, engine assigns; consent to rules; desk check-in (QR pass) and check-out (damage per item); door codes	Fairness; no e-signature; keys need Staff
Payment, communication	VietQR payment; in-app, email, SMS	VietQR transfer + claim confirmed by a Manager; in-app, push, email	No bank API; SMS cost
Added scope	—	Term transition (stay length, requests window, next-Term fees), registration-period requests, late check-in, Round 2, Utility and debt	Iteration 1: a dorm runs across Terms
Integration, technology	SIS link; forecast; Flutter	Student list import + reconciliation; dashboards; Flutter mobile app	SIS not accessible

1.2 Out of Scope

Item	Reason
Room or Bed self-selection by students	The engine keeps gender and Block rules and fairness (D2)
Electronic contract signing	Needs a legal e-signature provider; consent is recorded instead
SMS notifications	Cost; in-app, push and email cover the cases
Live SIS integration, bank API, forecasting, iOS app	No access or contract; list import, reconciliation, Manager confirmation, reports and the Flutter app (Android) instead

2. System at a Glance

FPT University's Tân Uyên campus will open dormitories for thousands of students and has no system to run them. CampusStay FPT takes a student from sign-in to a Bed and through every Term of the stay: Applications by Room Type and Room Class, a fair batch assignment, VietQR payments confirmed by Managers, door codes, the check-in and check-out desk, Term transitions for continuing residents, maintenance with time limits, monthly Utility billing and reports. Users: students (web and the Flutter mobile app), Staff of each Block, dormitory Managers and the Administrator. Core value: every Bed, payment and deadline follows one set of configurable

rules, with an audit trail, instead of spreadsheets and messages. Architecture: one Spring Boot API with PostgreSQL, a React web app and a Flutter mobile app built for Android (Report 4 §1.1, Figure 1).

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	System structure	Microservices; modular monolith	One Spring Boot API + one PostgreSQL, layered packages per feature: Allocation and payment in one ACID transaction; one API for three clients	Lê Thành Minh Trung	SDD §1.1–1.2
D2	How Beds are given	Pick a Bed first come; Staff by hand; batch engine	Students choose Room Type and Class; a deterministic engine places by priority and fit; a Manager approves per Block: fair, no race	Lê Thành Minh Trung	FR-ASG-01..03
D3	Renewal of residents	Re-apply each Term; auto-renew	Next-Term Allocation on the same Bed at M2; requests to M3, decisions by M4, fees M6, unpaid at M7 releases the Bed	Nguyễn Xuân Lộc	FR-ASG-07, 08; act-wf03
D4	Room Class and prices	Class per Floor; prices editable	Class per Room per Term; prices copied into the Term at publication and locked once Invoices exist	Trần Võ Quang Duy	FR-INV-02, FR-CFG-05
D5	Notification channels	SMS gateway; email only	In-app always, FCM push, email via an outbox in the same transaction: no SMS cost, nothing lost, two languages	Võ Minh Luân	FR-NTF-01, 02

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Dormitory not operating yet: real operation may differ	Rework after opening	Requirements from FPT's running dorms, Circular 27/2011, VNU-HCM; every price, window and rule configurable; weekly mentor review	Lê Thành Minh Trung
R2	Manual claim confirmation at the peak (≈ 3,000 Deposits in 48 h)	Students wait	Claim stops the clock; bulk confirm with count and total; hourly summary	Lê Thành Minh Trung
R3	One small VM and a Burstable database	Slow or down at peak	NFR-01 load test; health check and rollback by image tag; bigger VM in the runbook	Võ Minh Luân

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G1	Registration v1 items not built as written (self-selection, e-contract, self check-in/out, SMS, forecast, live SIS)	Registration form	Form v2 changes them (BM03)	Lê Thành Minh Trung	31/10/2026
G2	Workflow tests TC-WF-01..05 are manual (SR-21)	Report 5 §4.2	Run on staging; automate in CI	Võ Minh Luân	30/11/2026
G3	NFR-01, 04, 06 not measured yet (SR-06)	Report 5 §4.4	Load script, uptime check, run times	Võ Minh Luân	15/11/2026
G4	No concurrency test for one Allocation per Bed per Term (SR-06)	TC-NFR-03	Parallel-commit test	Nguyễn Xuân Lộc	15/11/2026
G5	Mobile app has only a few automated tests and no manual test cases (SR-08)	mobile/, Report 5	UI tests of key screens	Võ Minh Luân	30/11/2026
G6	Statuses in the schema never stored (SR-12); Term never Closed (SR-13)	V4, V6, V7; SDD §3.1	Drop or use them; close Terms after the stay	Lê Thành Minh Trung	15/11/2026

ID	Gap	Location	Plan	Owner	Due
G7	Round 2 run by a Manager not built; freed Beds not offered (SR-01, High)	FR-APP-07	Closed 28/10/2026: Manager › Assignment › Round 2 (TC-APP-024)	Lê Thành Minh Trung	28/10/2026

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Sign-in (password, Google)	All	Valid credentials give a session; the 5th wrong password locks the account; a new @fpt.edu.vn Google account gets no password until it sets one; no token → 401, wrong role → 403	TC-F01-01..04	Done
F-02	Student accounts: verification and list import	Student, Manager	A pending account cannot apply; verify/reject notifies the student; an import previews each line and saves only valid lines	TC-F02-01..06	Done
F-03	Registration-period request and Term publication	Manager, Admin	A 2nd open request for a Term is refused; milestones out of order are named per field; publishing fixes the prices	TC-F03-01..05	Done
F-04	Apply for a Term	Student	Submit creates an Awaiting Deposit Application and a Deposit = Deposit % × Term price; a 2nd Application or stay length above the maximum is refused	TC-F04-01..05	Done
F-05	VietQR payment claim and confirmation	Student, Manager	A claim stops the deadline; a rejection needs a reason and returns the time left (≥ 12 h); a Paid or Void Invoice cannot be claimed; refunds go to the student's saved bank account	TC-F05-01..04	Done
F-06	Review, assignment and Round 2	Manager, Student, System	A rejection needs a note; the same input gives the same proposal; commit creates Allocations and Balances; Staff cannot run; Round 2 offers free Beds only to waitlisted students without an open offer; withdrawing, or no Bed by check-in, cancels with the whole Deposit refunded	TC-F06-01..08	Done
F-07	Balance, forfeit and door code release	Student, Staff, System	A Balance unpaid at its deadline is forfeited; the code is released only when paid, after check-in start and with a confirmed code	TC-F07-01..05	Done
F-08	Check-in and check-out desk	Staff	Check-in is refused until the Balance is paid; check-out deducts Utility share and damage per item from the Utility Deposit; final readings entered per Room reach every leaver's draft	TC-F08-01..05	Done
F-09	Late check-in	Student, Staff, Manager	A request after check-in end + max days is refused; approval moves the Balance deadline; rejection needs a reason	TC-F09-01..03	Done
F-10	Next-Term requests	Student, Staff	An approved change moves the next-Term Bed; a Room Class change needs all continuing roommates; extension without debt is automatic	TC-F10-01..06	Done
F-11	Requests window, reconciliation and next-Term fees	Manager, System	M2 creates next-Term Allocations once; M3 releases un-extended Beds; fees wait for decisions and reconciliation; M7 releases unpaid Beds	TC-F11-01..06	Done
F-12	Turnover and operational moves	Staff, Manager	A move reads both meters, charges nothing and asks a new code for the old Room; after fees Staff cannot move next-Term Beds	TC-F12-01..03	Done
F-13	Maintenance with routing, time limits and escalation	Student, Staff, Manager	A request gets the routed assignee and the Block deadline (urgent < normal); other Blocks' Staff do not see it; unassigned ones escalate; only the person in charge changes its status, others take it over first	TC-F13-01..07	Done

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-14	Meter readings and Utility Invoices	Staff	A reading below the previous is refused; a CSV with a bad line saves nothing; closing issues Invoices split by days, rounded up	TC-F14-01..04	Done
F-15	Utility debt and termination	System, Manager	One overdue Utility Invoice gives warning 1 once; two overdue months after the final warning terminate the stay; past move-out it is Overdue	TC-F15-01..04	Done
F-16	Inventory, Room Class by Term, furniture	Manager, Staff	A Room whose Beds are given keeps its class; only Managers set classes; moving an item updates conformance	TC-F16-01..04	Done
F-17	Notifications, announcements and push	All	Staff announce only to their Blocks; pinned announcements top the board; push after commit, dead tokens disabled	TC-F17-01..04	Done
F-18	Dashboards, reports, exports and search	Staff, Manager, Admin	Exports are CSV with the screen's columns and audited; search stays in the user's Blocks; Term dashboard shows milestones	TC-F18-01..04	Done

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	Performance	p95 of 10 main read endpoints, 200 users, base data	≤ 1 s; errors < 1 %	Load script on a local server with production settings
NFR-02	Security	Permission and scope tests passing	100 %	Integration tests in CI
NFR-03	Integrity	Duplicate Allocations of a Bed in a Term	0	Unique indexes + concurrency test
NFR-04	Availability	Health checks answering UP	≥ 99 % from 01/11/2026	Uptime monitor, every 5 min
NFR-05	Usability	Texts with a Vietnamese translation	100 %	MessagesTest + scan of web routes
NFR-06	Deployability	Push to healthy production; rollback	≤ 15 min; ≤ 5 min	Actions run times; timed rollback

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Lê Thành Minh Trung (SE183752)	Leader, PM, BA, full-stack	Applications and payments (APP, BIL claims); assignment and residence (ASG); architecture and DevOps	WF-01, WF-05	D1, D2; ADR-17, ADR-22; v2 phases 1–8
Nguyễn Xuân Lộc (SE183782)	Backend developer	Term transition (requests window, next-Term requests, fees); enrollment reconciliation	WF-03	D3; v2 phase 6
Trần Võ Quang Duy (SE182146)	Frontend developer (web)	Registration periods and the Term wizard (CFG + web); inventory, furniture and door-code screens; web UI and i18n	WF-02	D4; v2 phases 3–4
Võ Minh Luân (SE182794)	Mobile developer, tester	Flutter mobile app and push (NTF); maintenance, routing and escalations (MNT, REQ); test documentation	WF-04	D5; mobile app

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Registration to residence	Student, Manager, Staff, System	A student signs in during an open registration period	The student is a resident with a door code	F-01, F-04, F-05, F-06, F-07	Application: Awaiting Deposit → Submitted → Approved → Allocated; Allocation: Assigned – unpaid → Paid → Resident; Invoice: Unpaid → Awaiting confirmation → Paid	Lê Thành Minh Trung
WF-02	Opening a registration period	Manager, Administrator, Student, System	A Manager requests a registration period	A published Term open to students	F-03, F-16, F-17	Registration-period request: Submitted → In progress → Done; Term: Draft → Published	Trần Võ Quang Duy
WF-03	Term transition of continuing residents	Student, Staff, Manager, System	M2 of the next Term	Continuing residents hold a paid Bed of the next Term; leavers check out	F-10, F-11, F-12, F-08	Next-Term Allocation: Continuing → Resident or Cancelled; Room-Change Request: Submitted → Approved	Nguyễn Xuân Lộc
WF-04	Maintenance with routing and time limits	Student, Staff, Manager, System	A resident reports a problem	The problem is fixed and confirmed within the time limit, or escalated	F-13, F-17, F-16	Maintenance Request: Submitted → Assigned → In Progress → Resolved → Closed	Võ Minh Luân
WF-05	Monthly Utility, debt and termination	Staff, Student, Manager, System	The month ends	Utility Invoices issued and collected; debtors warned or terminated	F-14, F-15, F-05, F-08	Invoice: Unpaid → Overdue → Paid; Termination: Notice sent → Checked out / Overdue	Lê Thành Minh Trung

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	WF-01	Main	Stage A (registration open); se220001 eligible, no Application	1. se220001 applies (4-bed, Standard, stay length 2) and claims the Deposit transfer. 2. manager.a confirms the claim and approves the Application. 3. Stage B: se220004 (assigned, unpaid) claims the Balance; manager.a confirms. 4. staff.a1 confirms the Room code; the release job runs.	Application Submitted then Approved; se220004 is a resident (orange Bed) and sees the code; each step notified	Lê Thành Minh Trung
DS-02	WF-01	Exception	Stage A; se220003 has one rejected Deposit claim	1. manager.a rejects se220003's second claim with a reason.	The Deposit Invoice is Void and the Application Expired; the student is told the reason; the Bed count is unchanged	Lê Thành Minh Trung
DS-03	WF-02	Main	Stage 0; manager.a, admin	1. manager.a requests a period for the next Term and imports the excluded Rooms from the sample	Request Done and linked to the Term; manager.a sees the changed value;	Trần Võ Quang Duy

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
				file. 2. admin builds the Term from the request, changes one price with a reason, completes the 7 steps and publishes.	students see the registration card and the announcement	
DS-04	WF-02	Exception	Stage 0; a request for the same Term is open	1. manager.b files a second request for the same Term. 2. admin enters M3 before M2 in the Timeline step.	The second request is refused; the Timeline step names the wrong milestone; nothing is published	Trần Võ Quang Duy
DS-05	WF-03	Main	Stage A (requests window open); se210006 in A-505	1. se210006 asks for a room change. 2. staff.a1 approves it with A-603 (stage B). 3. manager.a uploads the enrolled list and resolves groups 1–2. 4. manager.a issues the next-Term fees; se210001 pays and claims.	The next-Term Allocation moves to A-603; fees issued once with count and total; se210001 is Continuing and paid	Nguyễn Xuân Lộc
DS-06	WF-03	Exception	Stage C; se200005 did not pay the next-Term fee by M7	1. The M7 job runs. 2. staff.a1 opens the turnover list.	The fee is Void, the next-Term Bed released, se200005 is on the leavers list with a check-out date	Nguyễn Xuân Lộc
DS-07	WF-04	Main	Stage A; se200009 in A-801, routing rules of Block A	1. se200009 reports an urgent air-conditioner fault with a photo and the item. 2. staff.a2 (routed) starts and resolves it. 3. se200009 confirms.	Assigned at once with the urgent deadline; Closed; the item keeps a repair entry; notifications at each step	Võ Minh Luân
DS-08	WF-04	Exception	Stage A; se200010 in A-802 has a request unassigned past its limit	1. The escalation job runs. 2. manager.a reassigns it.	One escalation in Overdue requests; after reassignment it closes once the request is back on track	Võ Minh Luân
DS-09	WF-05	Main	Stage A; staff.a1; meter sample file of Block A	1. staff.a1 imports the readings and closes the month. 2. se200001 has an overdue Utility Invoice; the overdue job runs. 3. se200001 claims the transfer; manager.a confirms.	One Utility Invoice per resident split by days; warning 1 sent once; after payment the warning closes	Lê Thành Minh Trung
DS-10	WF-05	Exception	Stage A; se200004 terminated for Utility debt (A-703)	1. The move-out date passes without check-out; the job runs.	Termination Overdue; Managers and Block Staff told; A-703 needs a new door code	Lê Thành Minh Trung

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	18	3	87
Main workflows (end-to-end)	5	1	5
State transitions of main entities	70	1	70
Agreed NFRs	6	1	6
Committed total			168

Automated in CI: 148 of 168 committed test cases (88 %); the rest are the end-to-end workflow cases, the manual state transitions and the measured NFRs (Report 5 §4).